

Histograms

In a histogram the AREA is the frequency, not the height. Height = frequency density = frequency $\div$ class width.



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THE TOPIC ON ONE MAP

Histograms

01 — Drawing histograms

class width $fd = f \div w$ bars touch

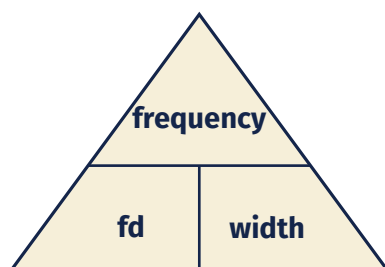
02 — Reading histograms

$f = fd \times w$ find the scale complete table

03 — Part of a bar

proportion probability estimate

THE TRIANGLE Cover the one you want:



frequency = $fd \times \text{width}$ (area of the bar)

$fd = \text{frequency} \div \text{width}$ (height of the bar)

no scale? count squares — area is still frequency

01 Frequency Density and Drawing Histograms

Add two columns to the table: class width and frequency density. Each bar spans its class exactly, bars touch, and the height is the frequency density.

WHEN YOU SEE → YOU DO

“draw a histogram”

→ width column, then $fd = f \div w$

unequal class widths

→ histogram, not a bar chart

“frequency density” on the axis

→ choose a scale that fits the biggest fd

“what mistakes did she make?”

→ heights equal to frequencies, gaps, wrong widths

WORKED MODEL • NOVEMBER 2021 P2H

$0 < L \leq 10$: 8, $10 < L \leq 15$: 13, $15 < L \leq 25$: 19, $25 < L \leq 40$: 24, $40 < L \leq 60$: 6.

1 widths 10, 5, 10, 15, 20

upper – lower

ANSWER $fd = 0.8, 2.6, 1.9, 1.6, 0.3$

× **TRAP** Plotting the frequency as the height. The height is the frequency density.

IDEA 01

Practise: Frequency Density and Drawing Histograms

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Frequency density

CH 6 • TOPIC 2 2 marks

Work out the frequency densities: $0 < t \leq 10$: 12, $10 < t \leq 15$: 20, $15 < t \leq 35$: 16.

SELECTED PRACTICE

Q2

PRACTICE

Practice • Medium

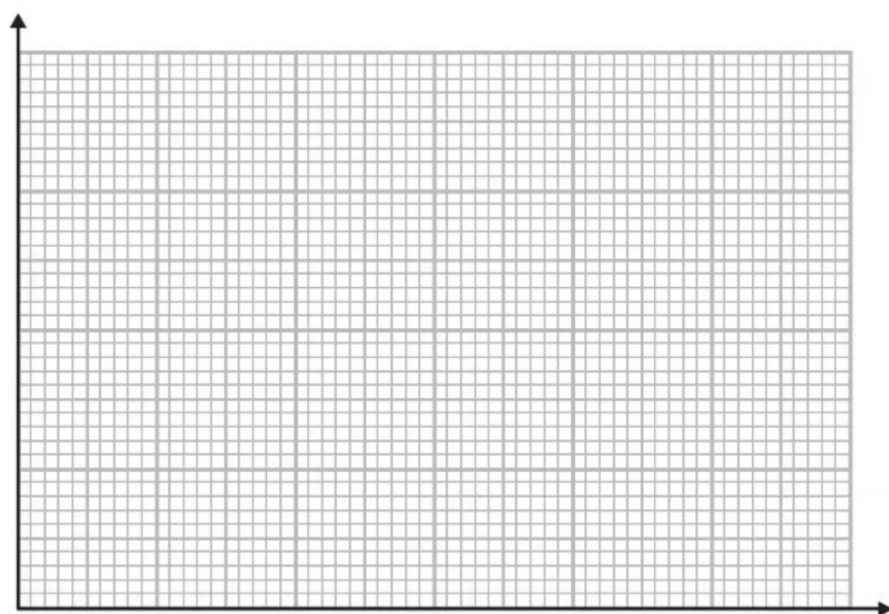
CH 6 • TOPIC 2

3 marks

- 1 The table gives information about the heights, h metres, of trees in a wood.

Height(h metres)	Frequency
$0 < h \leq 2$	7
$2 < h \leq 4$	14
$4 < h \leq 8$	18
$8 < h \leq 16$	24
$16 < h \leq 20$	10

Draw a histogram to show this information.



(3 marks)

Q3

PRACTICE

Practice • Medium

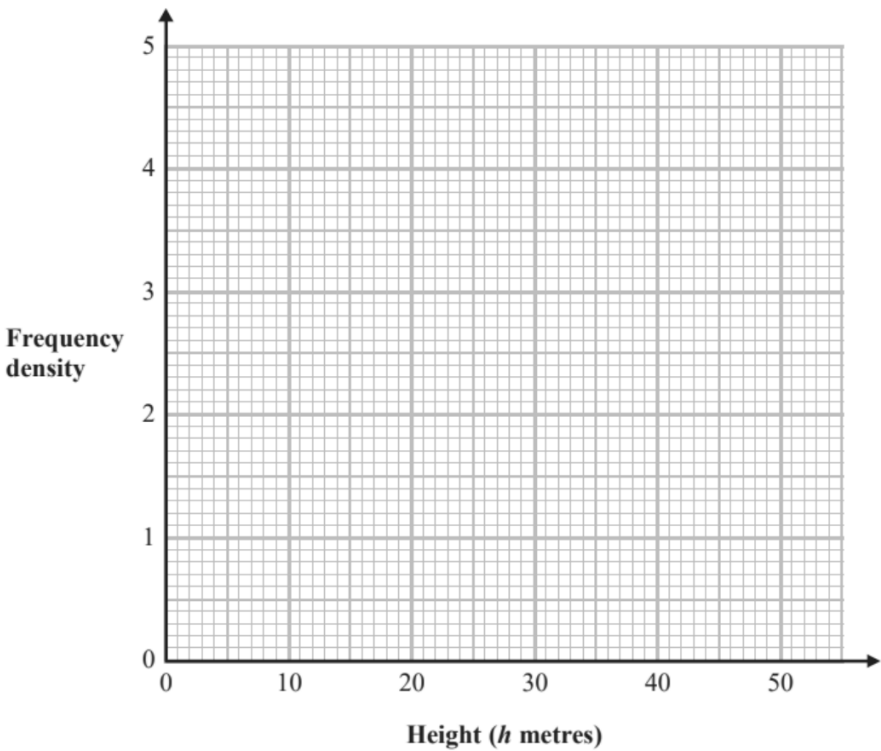
CH 6 • TOPIC 23 marks

PRACTISE • 6.2

8 The table gives information about the heights of some trees.

Height (h metres)	Frequency
$0 < h \leq 20$	15
$20 < h \leq 35$	48
$35 < h \leq 40$	21
$40 < h \leq 50$	16

On the grid, draw a histogram for this information.



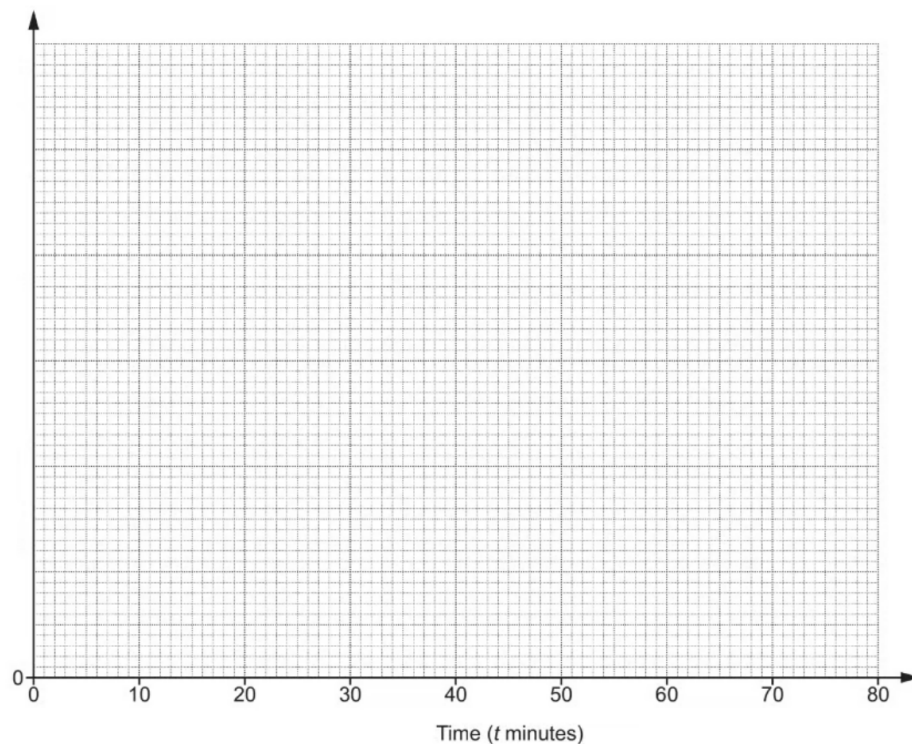
(3 marks)

Q4**PRACTICE** Practice • Medium**CH 6 • TOPIC 2** 4 marks

- 12 Ceri records the time taken, t minutes, to travel to school for a sample of 168 students at her Academy.

Time taken (t minutes)	Frequency
$0 < t \leq 10$	54
$10 < t \leq 20$	50
$20 < t \leq 40$	44
$40 < t \leq 80$	20

Draw a histogram to represent this information.

**(4 marks)**

Q5

PRACTICE

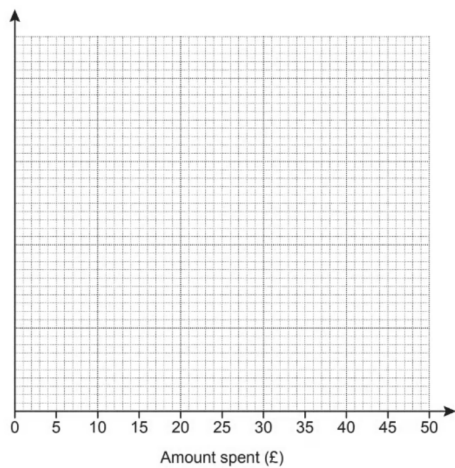
Practice • Very Hard

CH 6 • TOPIC 28 marks

17 (a) Ana records the amount of money spent by 140 customers in her shop on one day

Amount spent (£ <i>a</i>)	Frequency
$0 < a \leq 5$	35
$5 < a \leq 10$	42
$10 < a \leq 15$	20
$15 < a \leq 20$	18
$20 < a \leq 30$	14
$30 < a \leq 50$	11

Draw a histogram to represent this information.



(4 marks)

(b) Ana wants to offer a discount to the customers who spend the most money in her shop.

Voucher Save 10% when you spend more than £...

She wants to give the discount to approximately 25% of her customers.

Suggest a suitable amount of money for Ana to use on her voucher.

Iustify your decision.

(4 marks)

PAST PAPERS

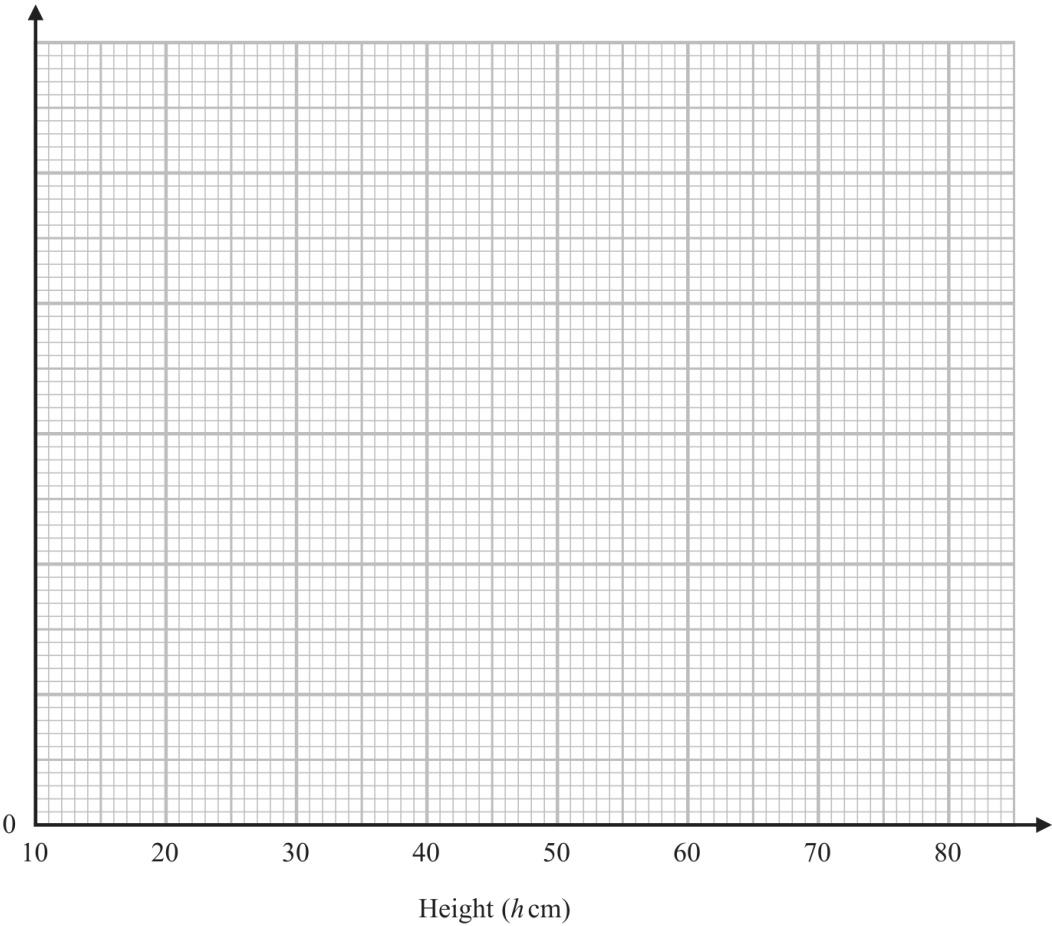
Q6 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q18

CH 6 • TOPIC 2 **5 marks**

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leq 20$	35
$20 < h \leq 35$	45
$35 < h \leq 50$	75
$50 < h \leq 70$	40
$70 < h \leq 80$	8

(a) On the grid, draw a histogram for this information.



(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

(Total for Question 18 is 5 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q14

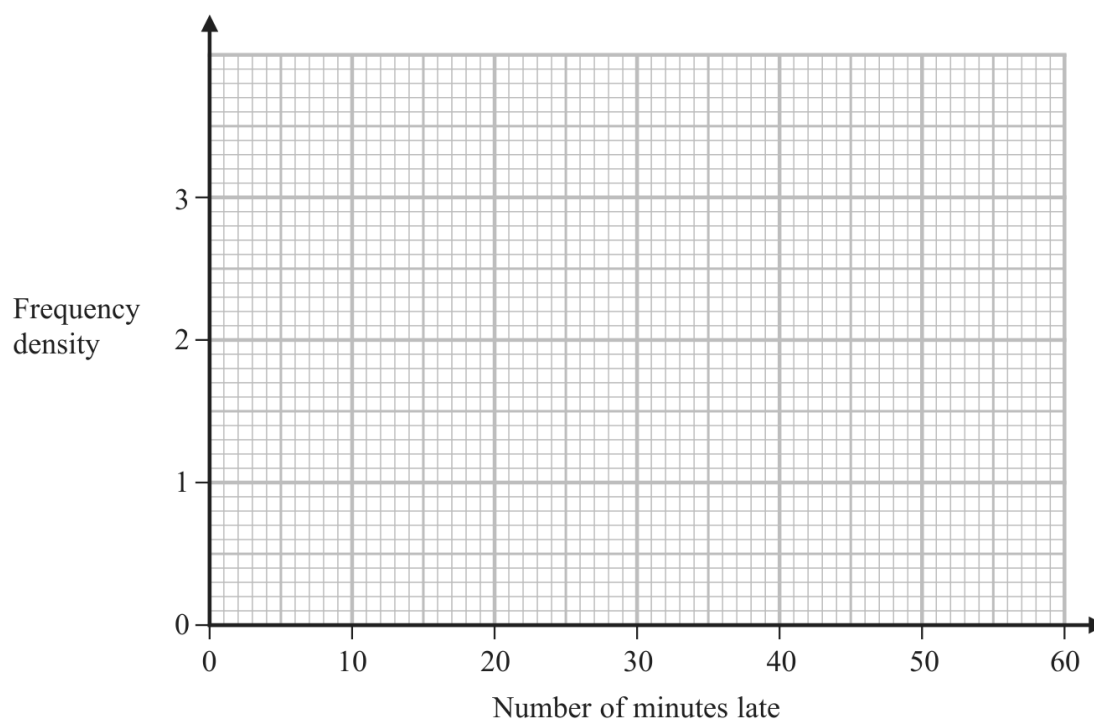
CH 6 • TOPIC 2 5 marks

14 Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leq 10$	8
$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

(Total for Question 14 is 5 marks)

Q8 **PAST PAPER** Edexcel 4MA1 • May 2024 • Paper 1HR • Q16

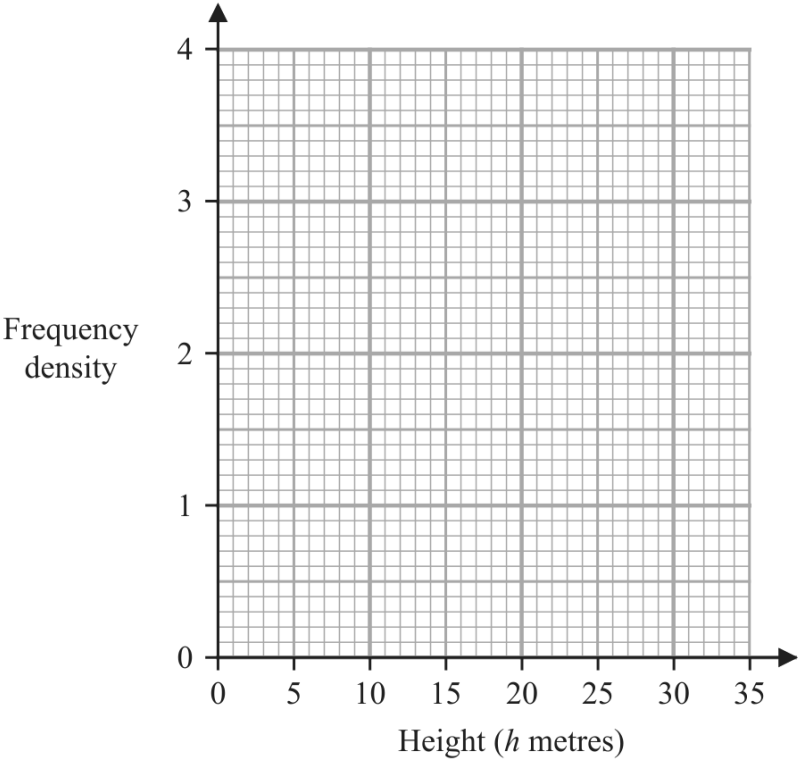
CH 6 • TOPIC 2 **3 marks**

PRACTISE • 6.2

16 The table gives information about the heights, in metres, of the trees in a park.

Height (h metres)	Frequency
$0 < h \leq 2$	5
$2 < h \leq 5$	12
$5 < h \leq 10$	18
$10 < h \leq 20$	14
$20 < h \leq 35$	9

On the grid, draw a histogram for this information.



(Total for Question 16 is 3 marks)

02 Reading a Histogram: Area is Frequency

When the axis has no numbers, use the bar you know: its area gives the value of one square. Then every other bar is just counting squares.

WHEN YOU SEE → YOU DO

“there are 160 students between 155 and 170”

→ area of that part = 160 → value of one square

a frequency density scale is given

→ $f = fd \times \text{width}$ for each bar

“complete the table and the histogram”

→ known bar → scale → both ways

“work out the total number”

→ add all the areas

“estimate the mean” from a histogram

→ frequencies first, then midpoints

Known bar

Value of one square (or fd scale)

Frequency of each bar

Answer

WORKED MODEL • JANUARY 2020 P1HR

Relative heights 18, 40, 20, 15, 6 over widths 15, 5, 10, 15, 5. $155 < h \leq 170$ has 160 students.

1

known part: $5(40) + 10(20) = 400$ area units = 160

scale

2

one unit = 0.4; total area = 925

all bars

ANSWER $925 \times 0.4 = 370$ students

IDEA 02

Practise: Reading a Histogram: Area is Frequency

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Area

CH 6 • TOPIC 2 2 marks

A bar goes from 20 to 50 with frequency density 1.4. How many values are in this class?

Q10 DR ESLAM Scale

CH 6 • TOPIC 2 3 marks

A bar from 10 to 15 is 8 small squares tall and holds 20 people. A bar from 15 to 30 is 4 small squares tall. How many people are in it?

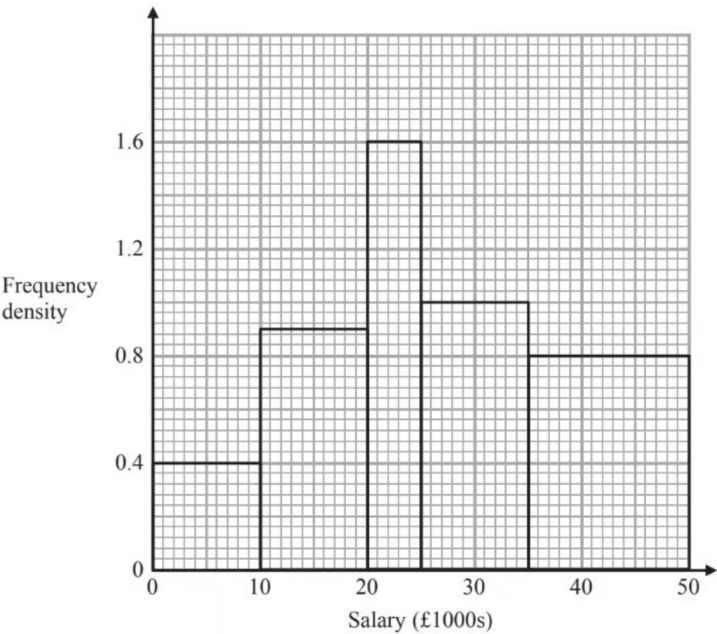
SELECTED PRACTICE

Q11 **PRACTICE** Practice • Hard

CH 6 • TOPIC 2 6 marks

PRACTISE • 6.2

1 (a) The histogram shows some information about the salaries of a sample of people.



Use the histogram to complete the frequency table.

Salary (<i>p</i>) in £1000s	Frequency
$0 < p \leq 10$	4
$10 < p \leq 20$	
$20 < p \leq 25$	
$25 < p \leq 35$	
$35 < p \leq 50$	

(2 marks)

(b) Work out the proportion of people in the sample who have a salary greater than £ 40 000.

(2 marks)

(c) Find an estimate for the median salary.

(2 marks)

Q12**PRACTICE**

Practice • Hard

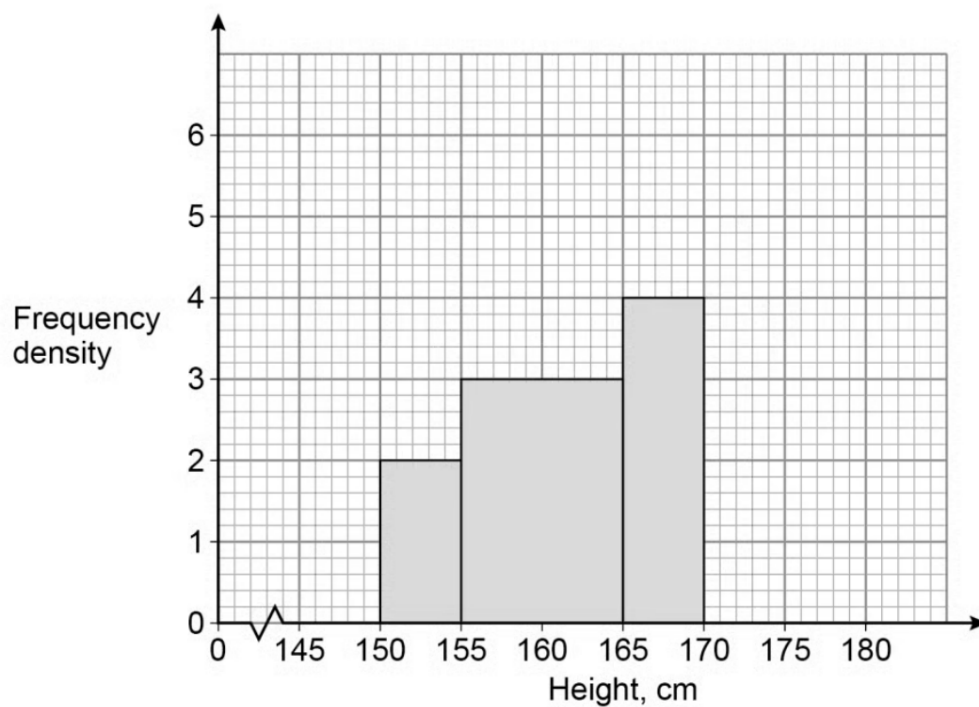
CH 6 • TOPIC 2

4 marks

12 A histogram is drawn to represent the heights of a sample of women.

Three of the four bars are shown.

The bar for $170 \text{ cm} \leq \text{height} < 180 \text{ cm}$ is missing.



There are 74 women in the sample.

Complete the histogram.

(4 marks)

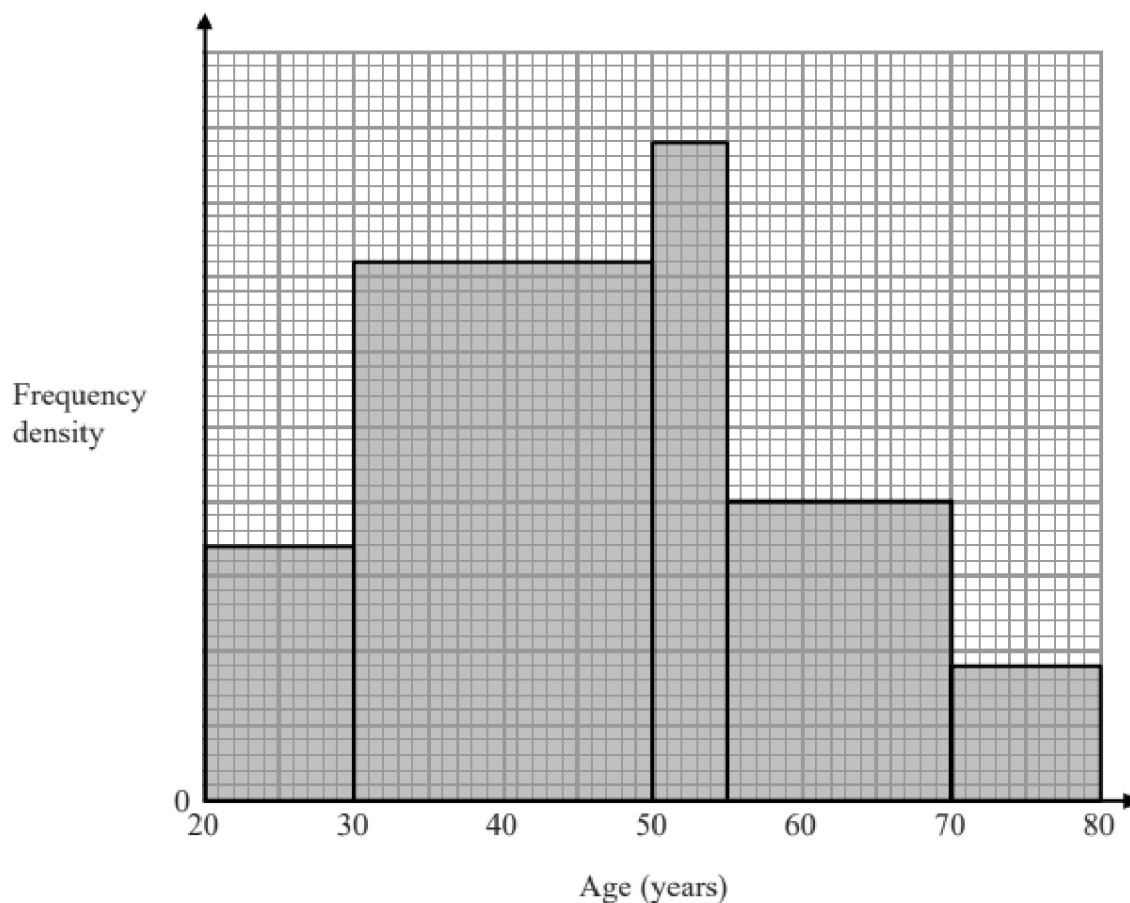
Q13**PRACTICE**

Practice · Hard

CH 6 • TOPIC 2 4 marks

18 150 people took part in a survey.

The histogram shows information about the ages of these people.



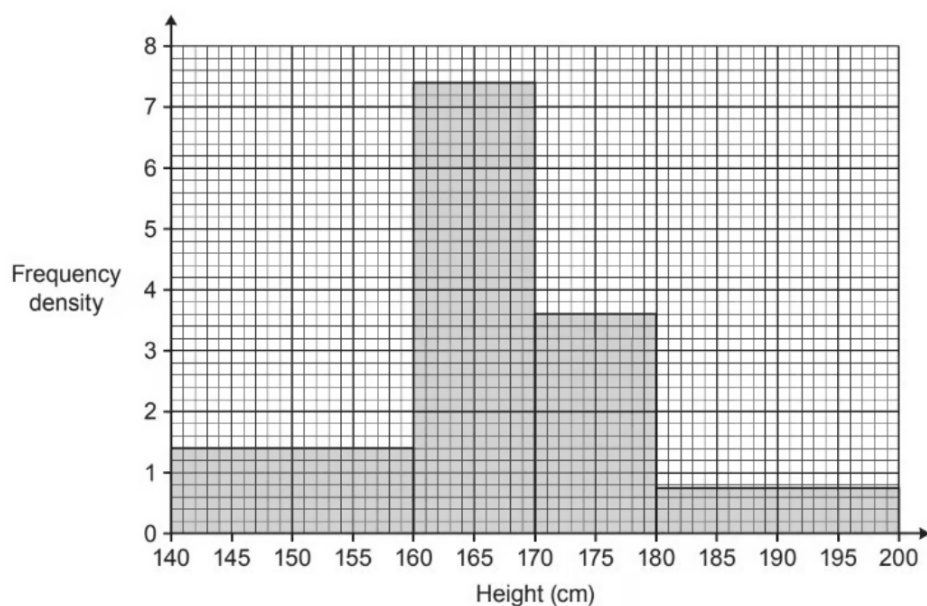
Work out how many of these 150 people are aged between 50 years and 55 years.

(4 marks)

Q14 PRACTICE Practice • Very Hard

CH 6 • TOPIC 2 5 marks

16 The histogram summarises the heights of the 153 members of a swimming club.



Members of a rowing club have a median height of 172 cm.

Which club has the greater median height? You must show all your working.

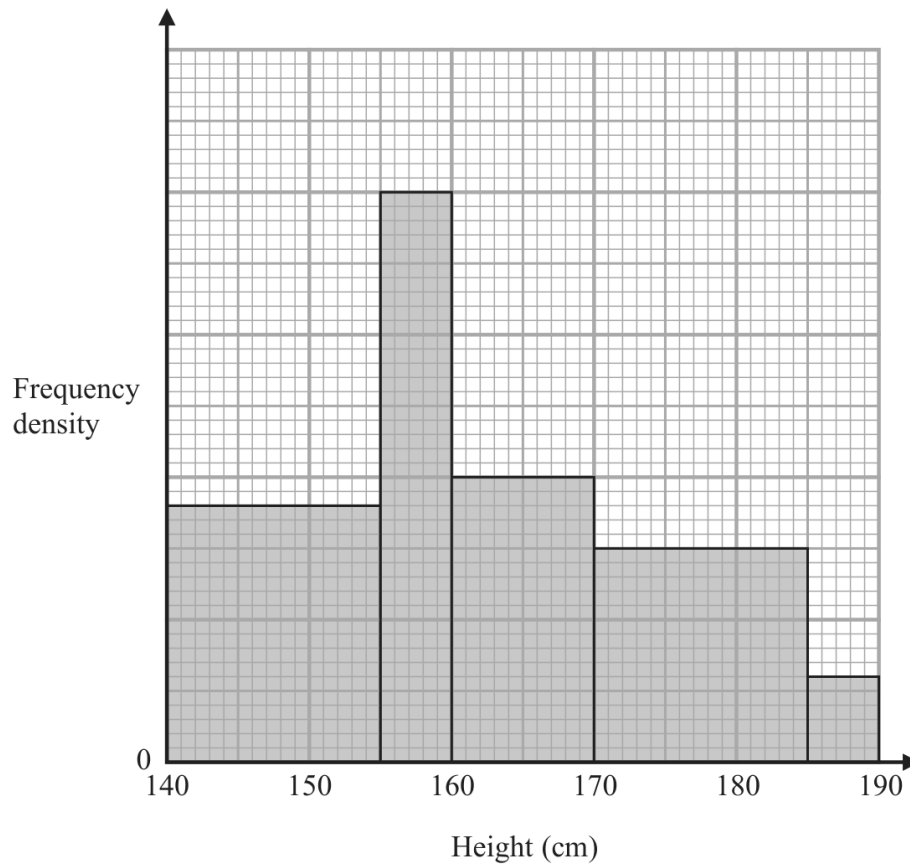
(5 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1HR • Q19

CH 6 • TOPIC 2 4 marks

19



The histogram gives information about the heights of all the Year 11 students at a school.

There are 160 students in Year 11 with a height between 155 cm and 170 cm.

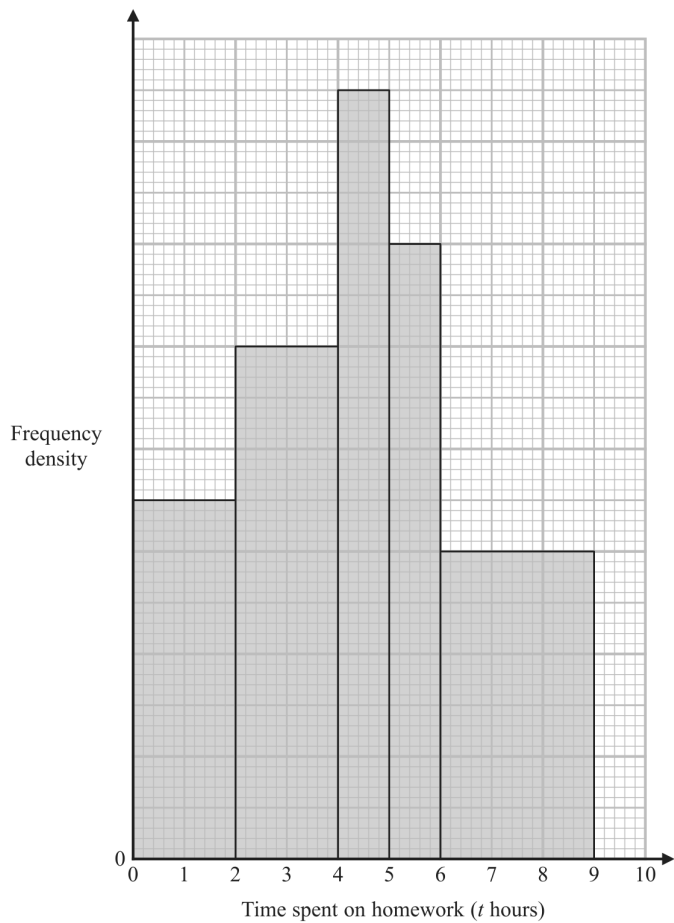
Work out the total number of students in Year 11 at the school.

(Total for Question 19 is 4 marks)

Q16 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 1HR • Q18

CH 6 • TOPIC 2 5 marks

18 The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week. No student in Year 11 spent longer than 9 hours on their homework.



Time spent on homework (t hours)	Frequency
$0 < t \leq 2$	28
$2 < t \leq 4$	
$4 < t \leq 5$	
$5 < t \leq 6$	
$6 < t \leq 9$	

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week. Give your answer in hours correct to 3 significant figures.

..... hours

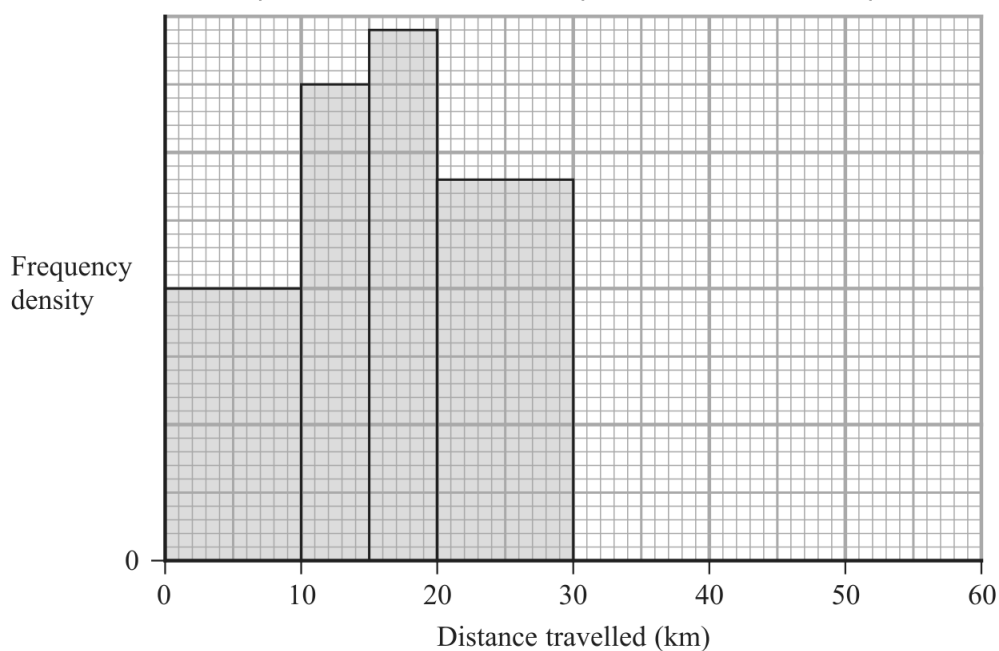
(Total for Question 18 is 5 marks)

Q17 PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q18

CH 6 • TOPIC 2 5 marks

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,
 (b) complete the histogram.

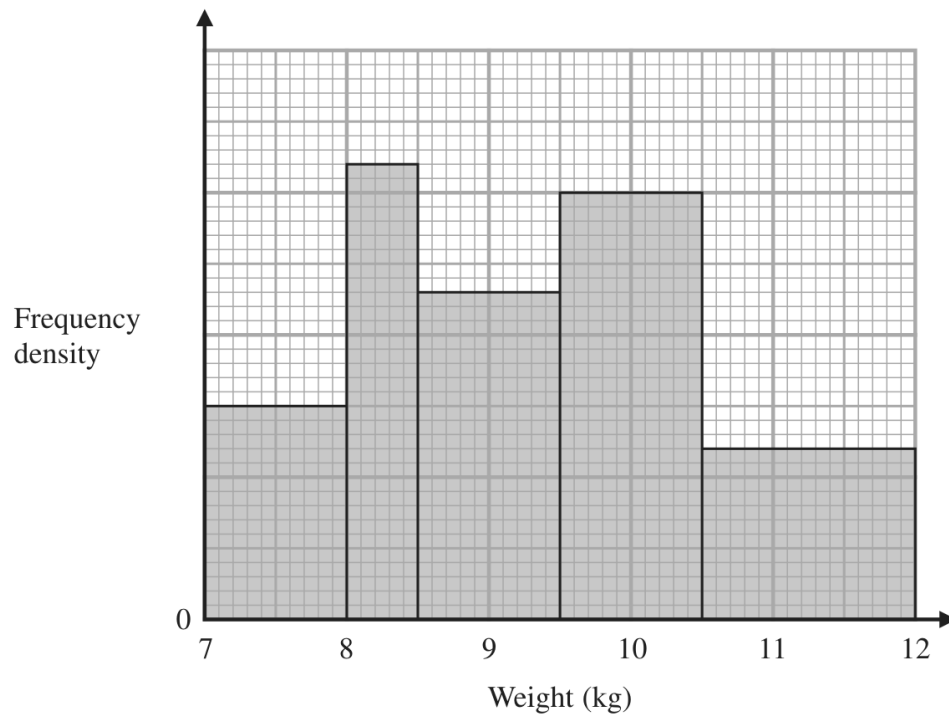
One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(Total for Question 18 is 5 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q21

CH 6 • TOPIC 2 3 marks**21**

The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

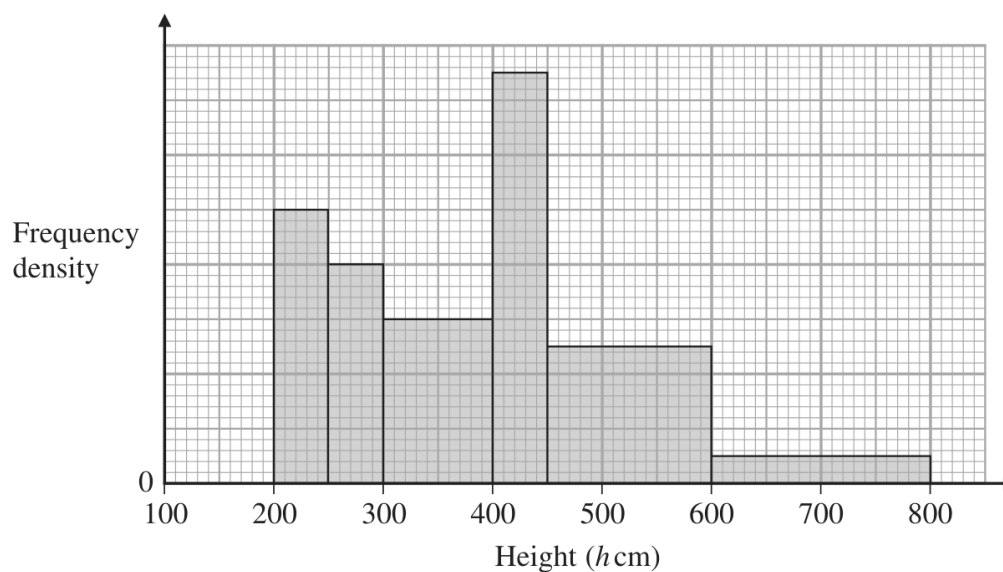
(Total for Question 21 is 3 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q19

CH 6 • TOPIC 2 3 marks

19 The histogram gives information about the height, h cm, of each tree in part of a forest.



There are no trees for which $h \leq 200$ and for which $h > 800$

The number of trees for which $300 < h \leq 400$ is 8 fewer than the number of trees for which $400 < h \leq 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

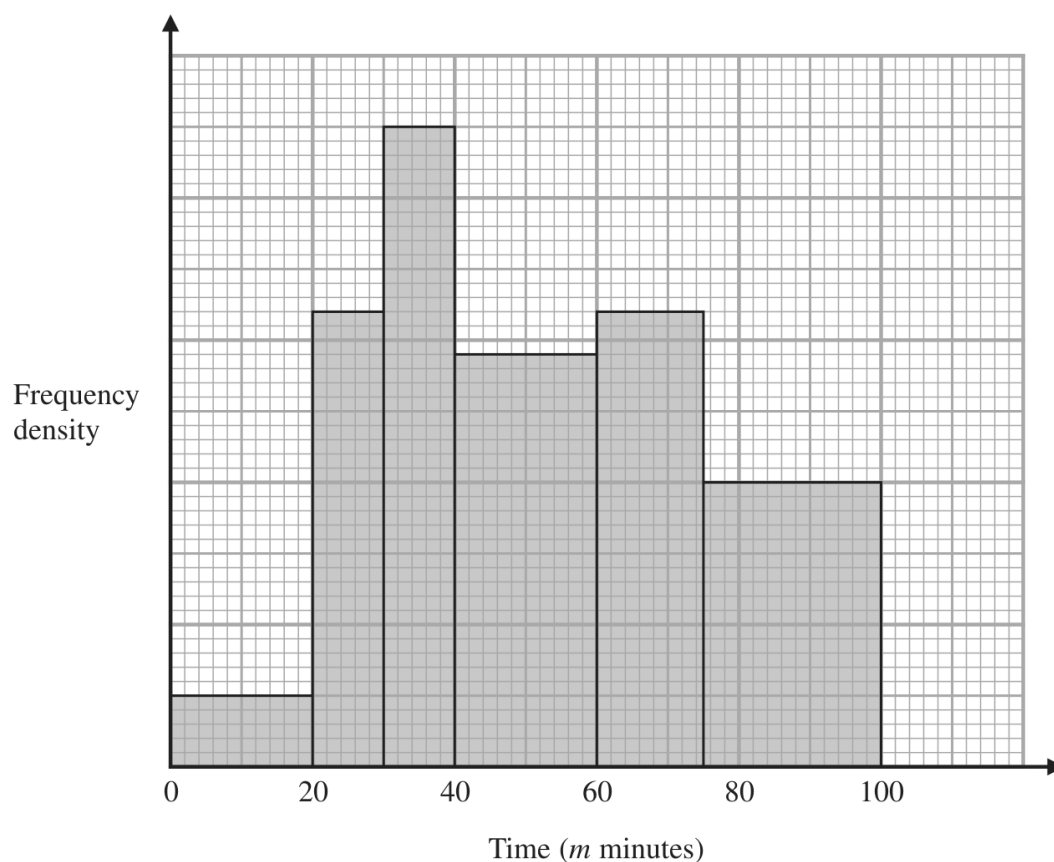
(Total for Question 19 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q21

CH 6 • TOPIC 2 3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

.....

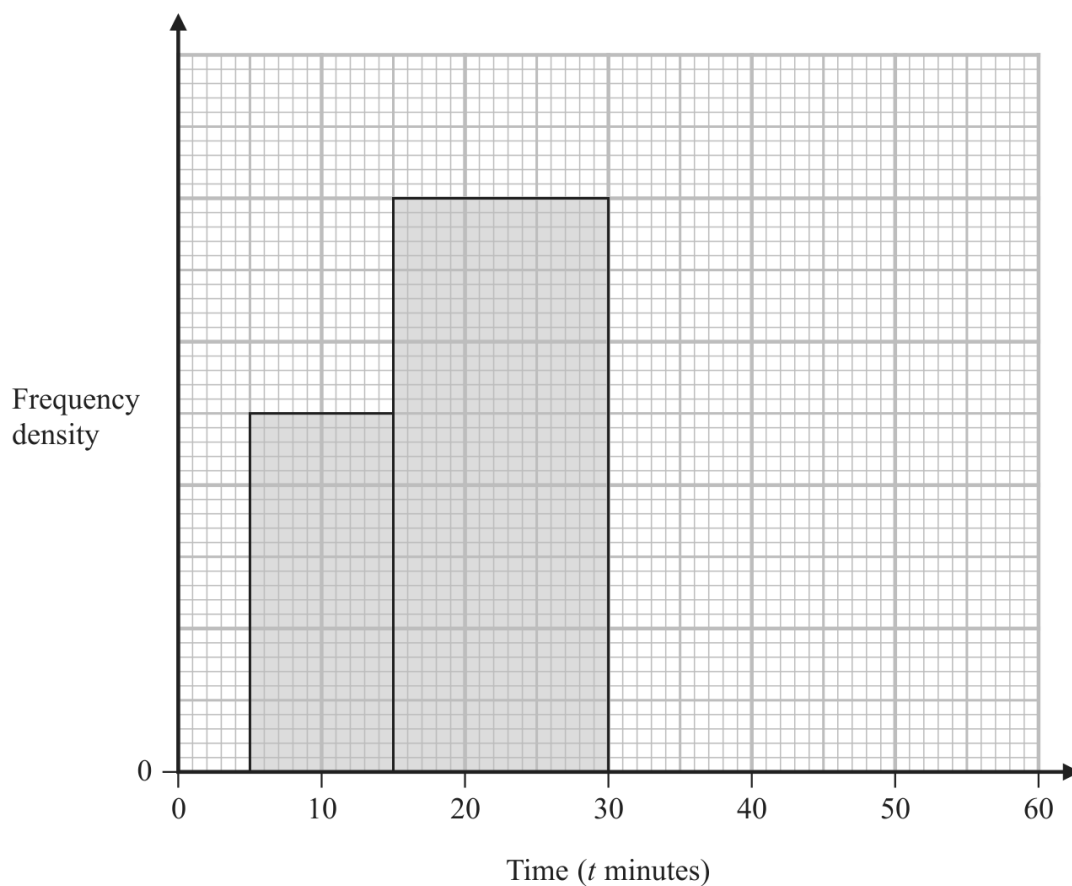
(Total for Question 21 is 3 marks)

Q21 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1H • Q18

CH 6 • TOPIC 2 4 marks

- 18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	Frequency
$0 < t \leq 5$	23
$5 < t \leq 15$	
$15 < t \leq 30$	
$30 < t \leq 40$	18
$40 < t \leq 60$	14



Complete the table and the histogram.

(Total for Question 18 is 4 marks)

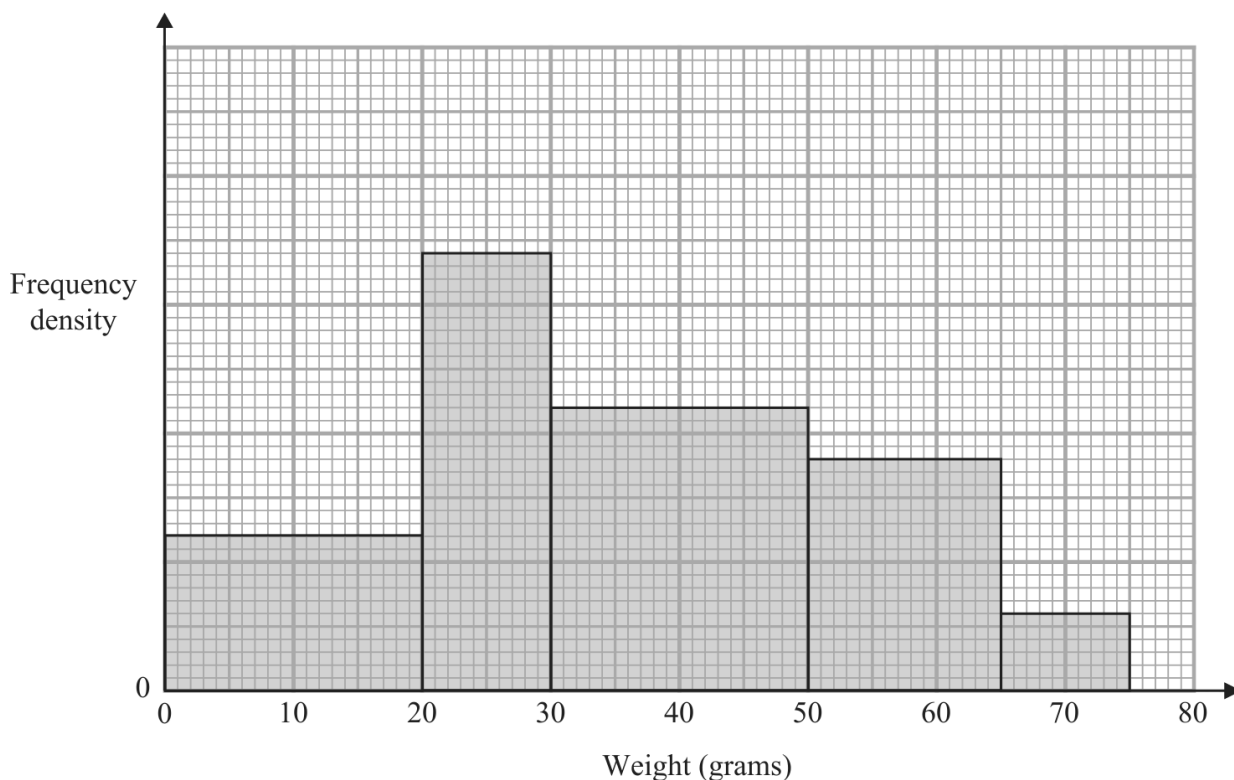
Q22

PAST PAPER

Edexcel 4MA1 • November 2023 • Paper 1H • Q17

CH 6 • TOPIC 2 3 marks

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

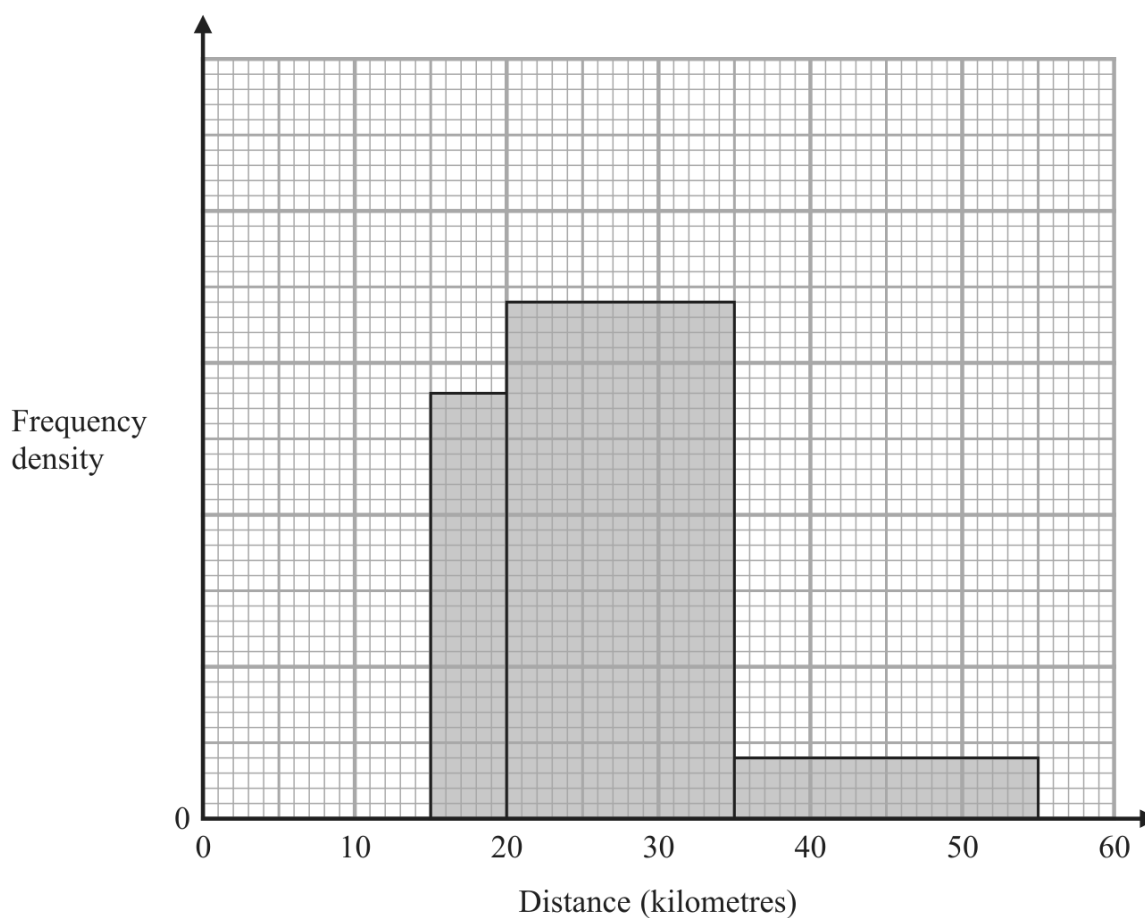
(Total for Question 17 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q22

CH 6 • TOPIC 2**3 marks**

- 22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.

None of the adults ran more than 55 kilometres.

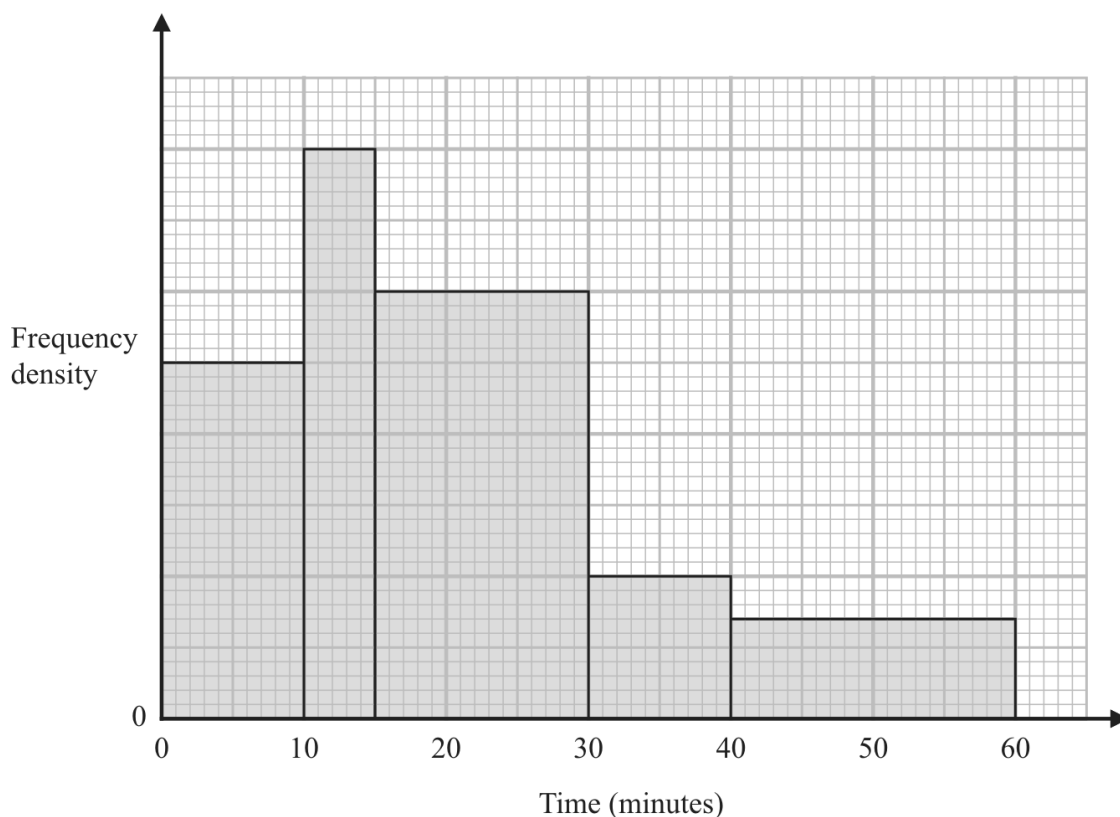
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • November 2025 • Paper 2H • Q21**CH 6 • TOPIC 2** **3 marks**

- 21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

03 Part of a Bar: Proportions and Probability

Inside a class, assume the values are spread evenly. Part of a bar = $fd \times$ the part of the width you need. Then divide by the total for a proportion or a probability.

WHEN YOU SEE → YOU DO

“between 17 and 35 minutes”

→ cut the end bars at 17 and at 35

“estimate the proportion / percentage”

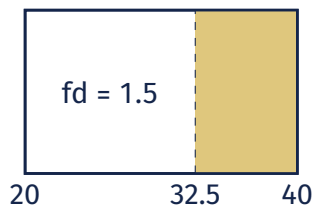
→ $\text{part} \div \text{total}$

“given that it spent more than 30 minutes”

→ the new total is only the bars above 30

“two parcels picked at random”

→ without replacement: $\frac{a}{n} \times \frac{a-1}{n-1}$



more than 32.5: $1.5 \times 7.5 = 11.25$

the cut is a fraction of the width

WORKED MODEL • JANUARY 2022 P2H

12 plants for $75 < h \leq 85$. Probability that a plant is taller than 82.5 cm?

1 scale: 20 small squares = 1 plant; total 600 squares = 30 plants

known bar

2 82.5–85: $\frac{1}{4}$ of 240 = 60; 85–90: 40 → 100 squares

part of a bar

ANSWER $\frac{100}{600} = \frac{1}{6}$

× **TRAP** Using the height instead of the area for part of a bar: always $fd \times \text{width}$.

IDEA 03

Practise: Part of a Bar: Proportions and Probability

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Part of a bar

CH 6 • TOPIC 2 2 marks

The class $20 < x \leq 40$ has frequency 30. Estimate how many values are between 20 and 25.

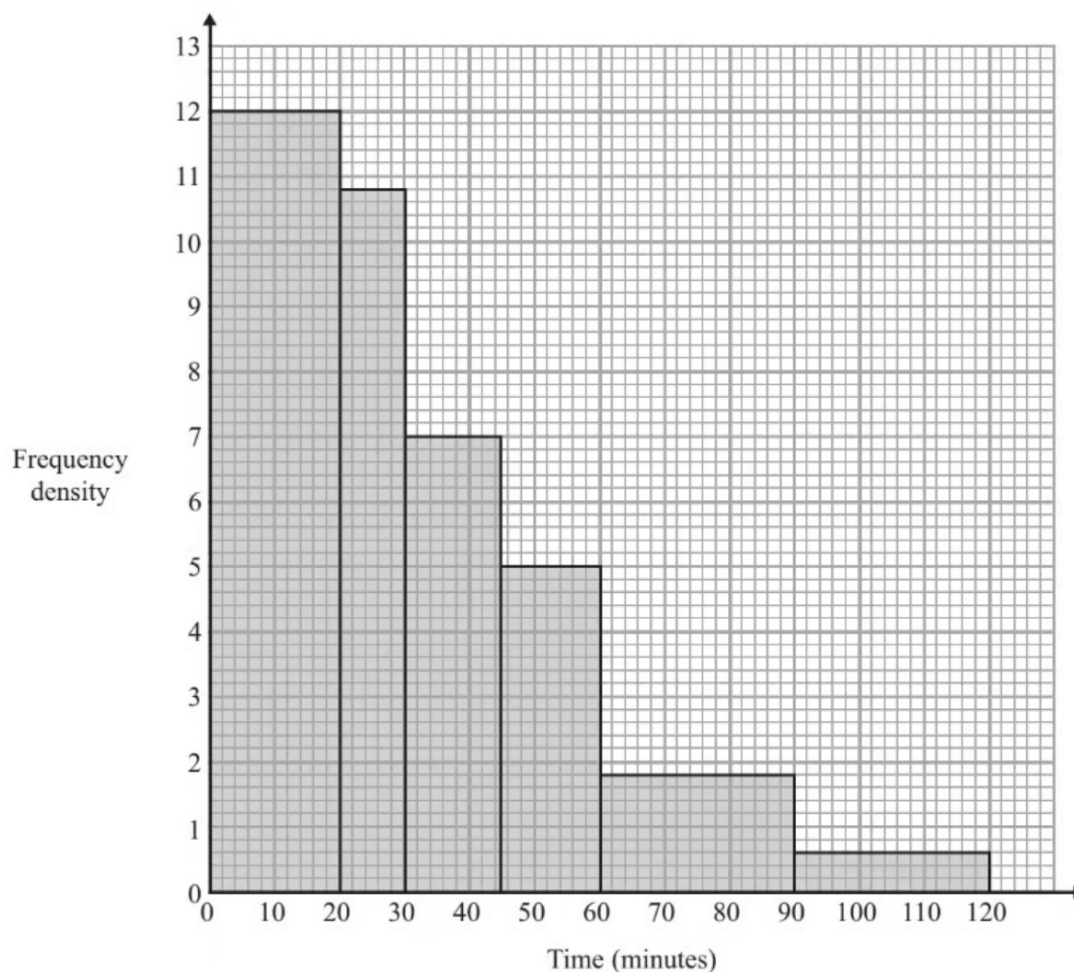
SELECTED PRACTICE

Q26**PRACTICE**

Practice • Hard

CH 6 • TOPIC 2 3 marks

- 8 The histogram shows information about the times, in minutes, that some passengers had to wait at an airport.



Work out the percentage of the passengers who had to wait for more than one hour.

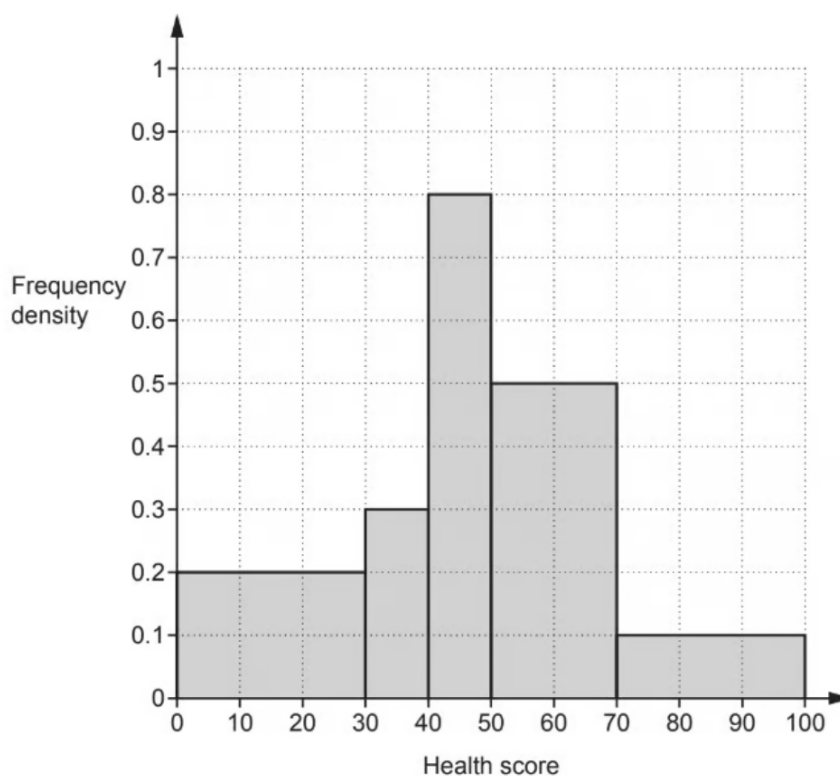
(3 marks)

Q27**PRACTICE**

Practice • Hard

CH 6 • TOPIC 2 5 marks

17 The histogram summarises a health score for a group of people.



i) Estimate the fraction of the group who had a score of 45 or more.

[4]

ii) What assumption did you make in answering part (i)?

[1]

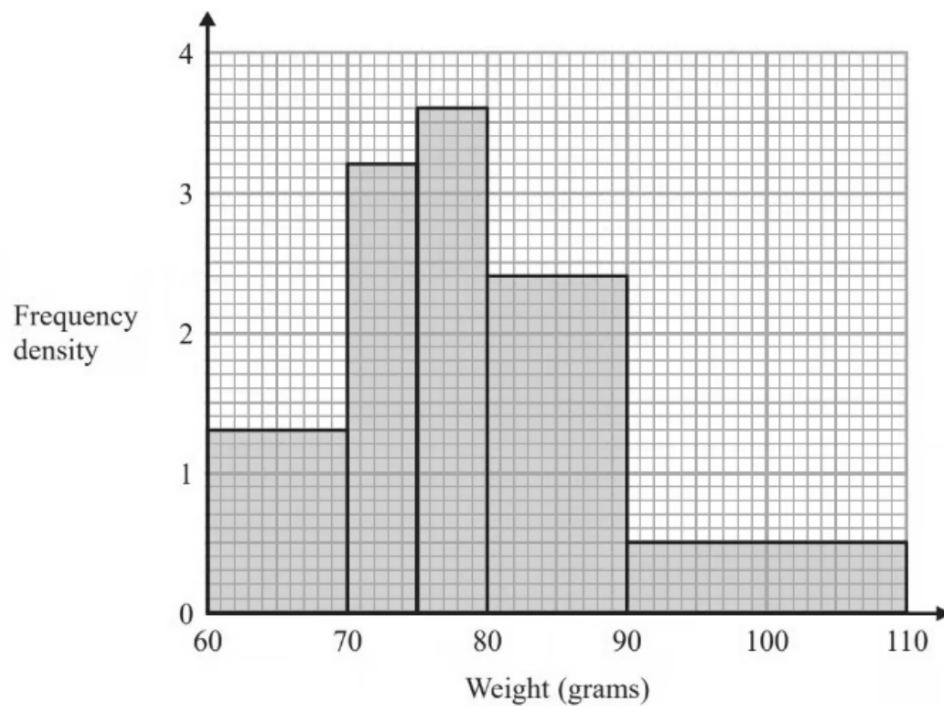
(5 marks)

Q28**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 2**3 marks**

- 1 The histogram shows information about the weights, in grams, of some plums.



Work out an estimate for the proportion of these plums with a weight of less than 100 grams.

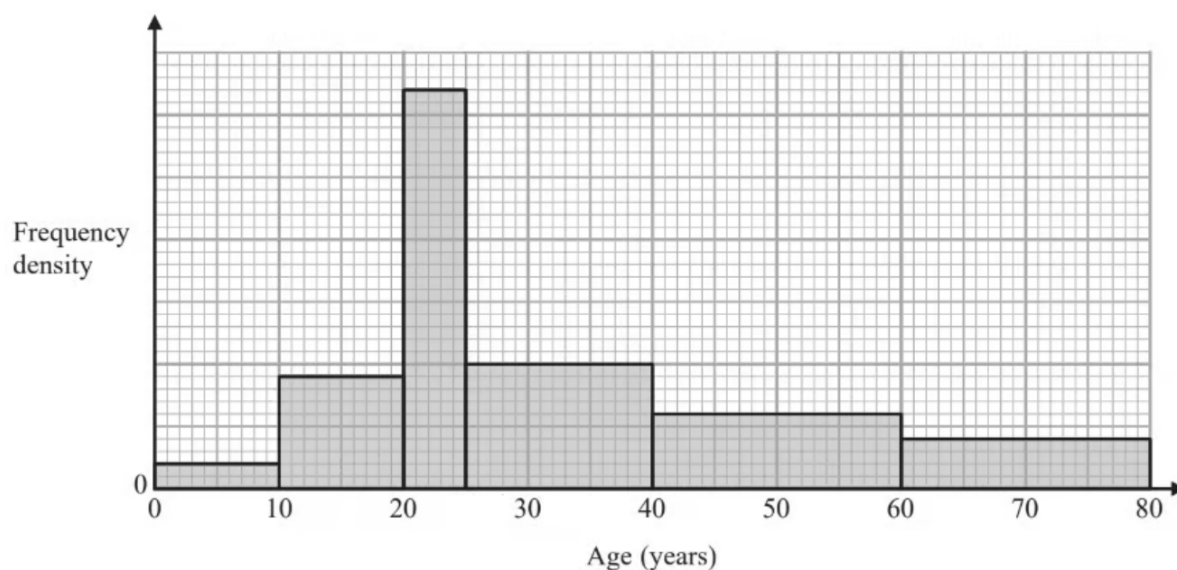
(3 marks)

Q29**PRACTICE**

Practice • Very Hard

CH 6 • TOPIC 2 **3 marks**

- 10** The histogram shows information about the ages of all the passengers travelling on a plane. No one on the plane is older than 80 years.



A passenger on the plane is picked at random.

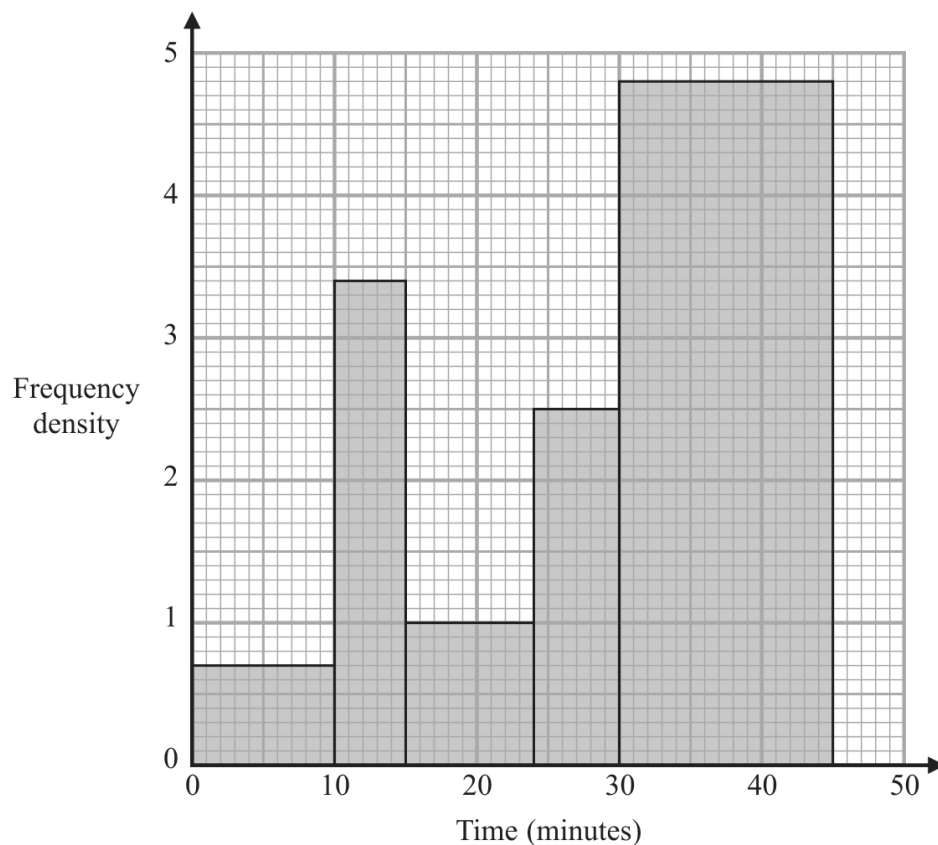
Work out an estimate for the probability that this person is older than 55 years.

(3 marks)

PAST PAPERS

Q30 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 2H • Q18**CH 6 • TOPIC 2** **5 marks**

- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



- (a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

.....
(3)

One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

.....
(Total for Question 18 is 5 marks)

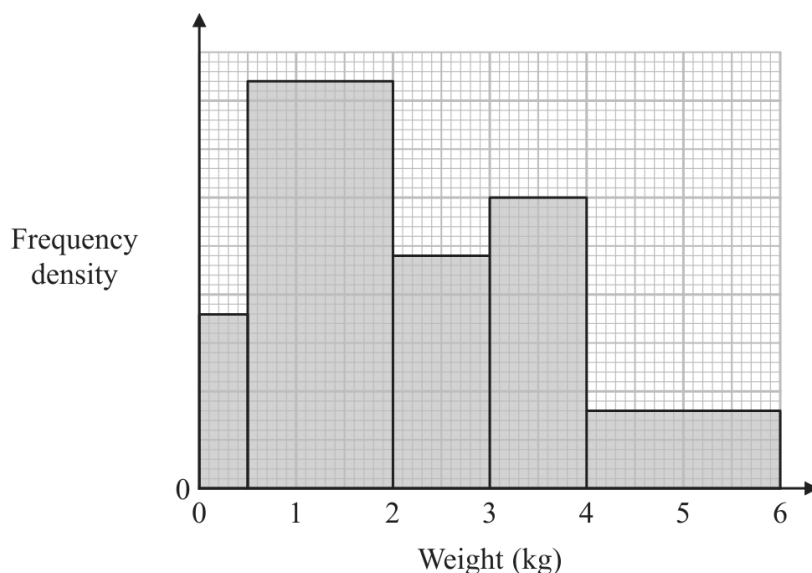
Q31**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q15

CH 6 • TOPIC 2 6 marks

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

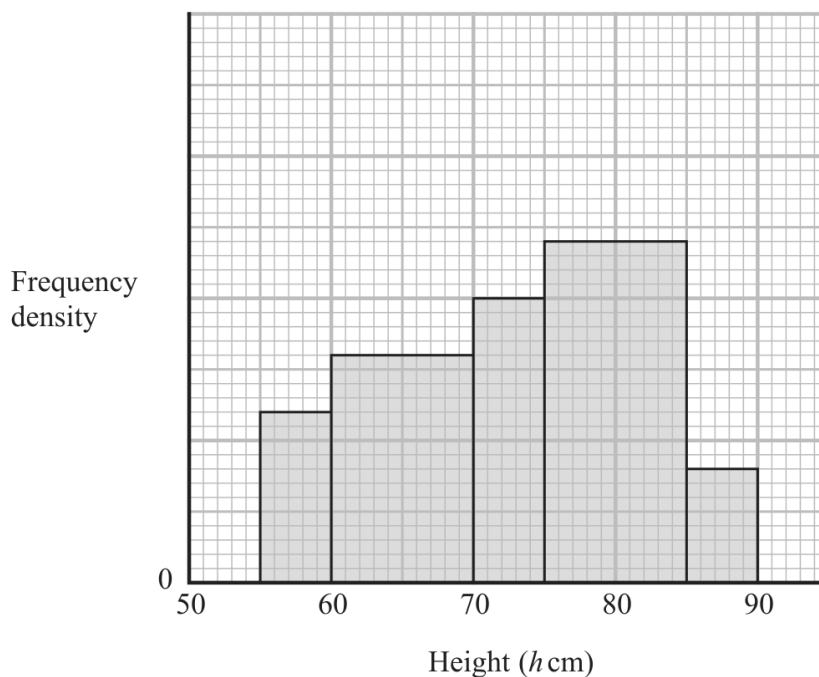
(Total for Question 15 is 6 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q20

CH 6 • TOPIC 2 4 marks

20 The histogram gives information about the heights, h cm, of some tomato plants.



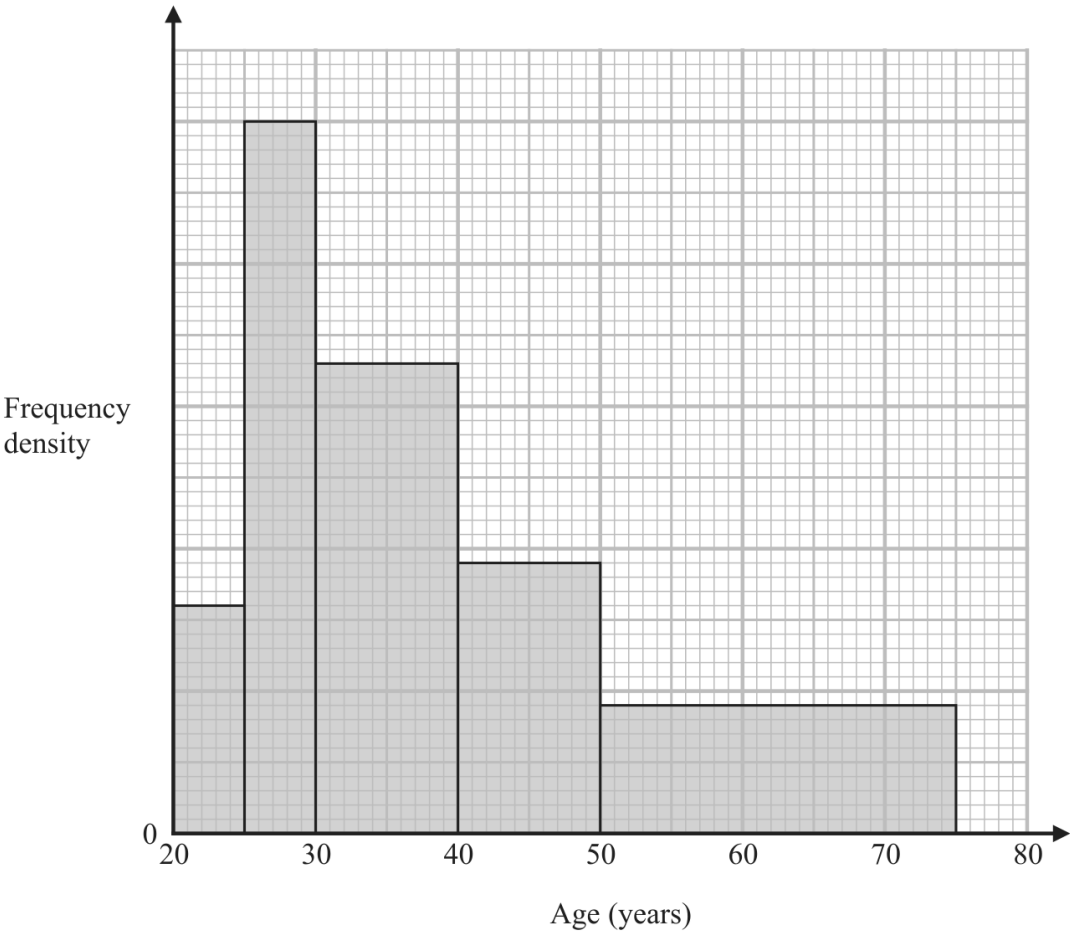
There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

22 Some people attend a concert.
The histogram shows information about the ages of these people.



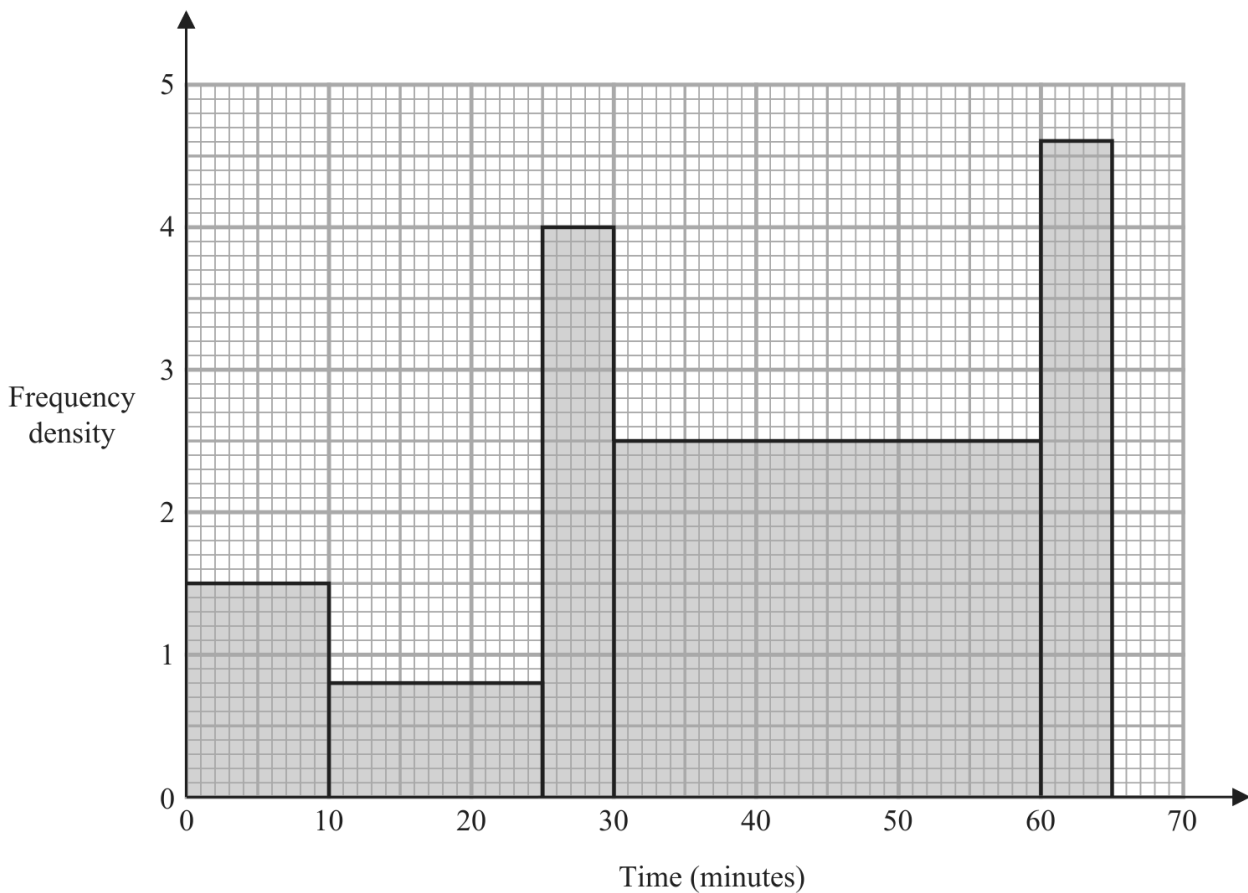
Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

.....%

(Total for Question 22 is 4 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2024 • Paper 1H • Q18**CH 6 • TOPIC 2** **4 marks**

- 18 The histogram shows information about the times some students took to complete a puzzle.



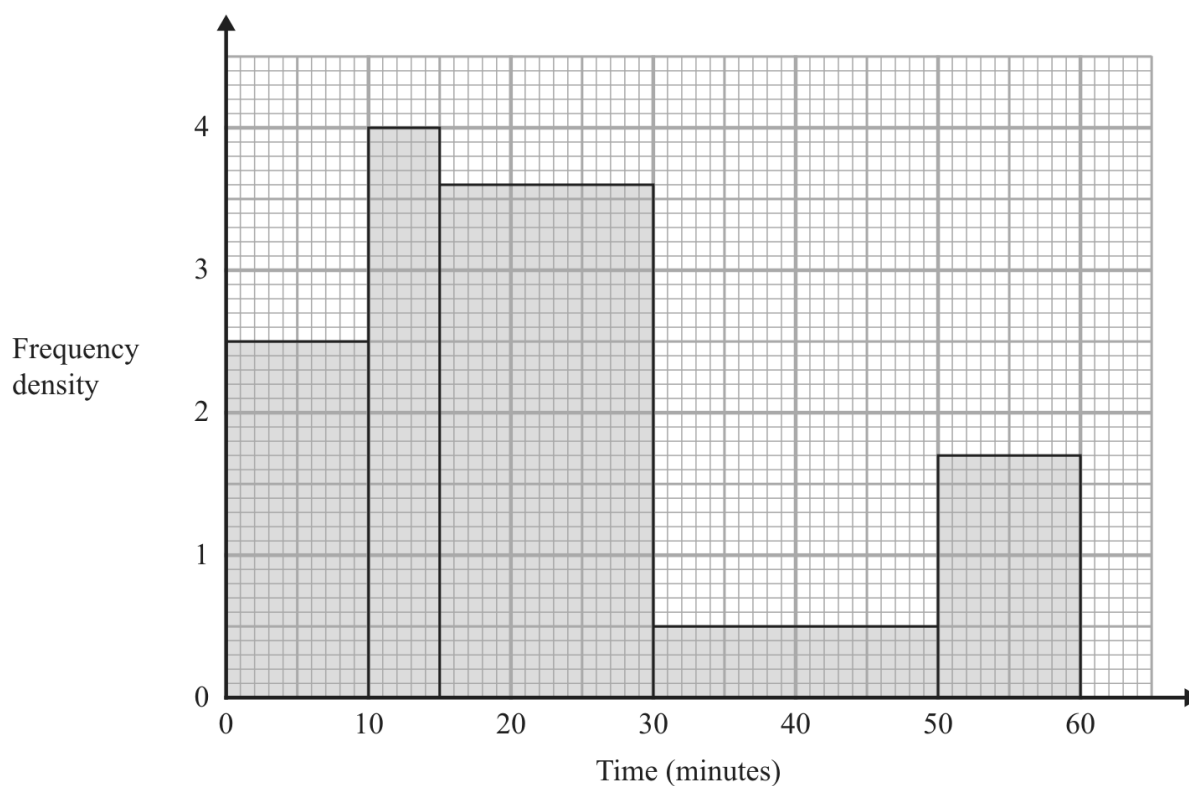
Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

.....

(Total for Question 18 is 4 marks)

Q35 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 1H • Q17**CH 6 • TOPIC 2** **3 marks**

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

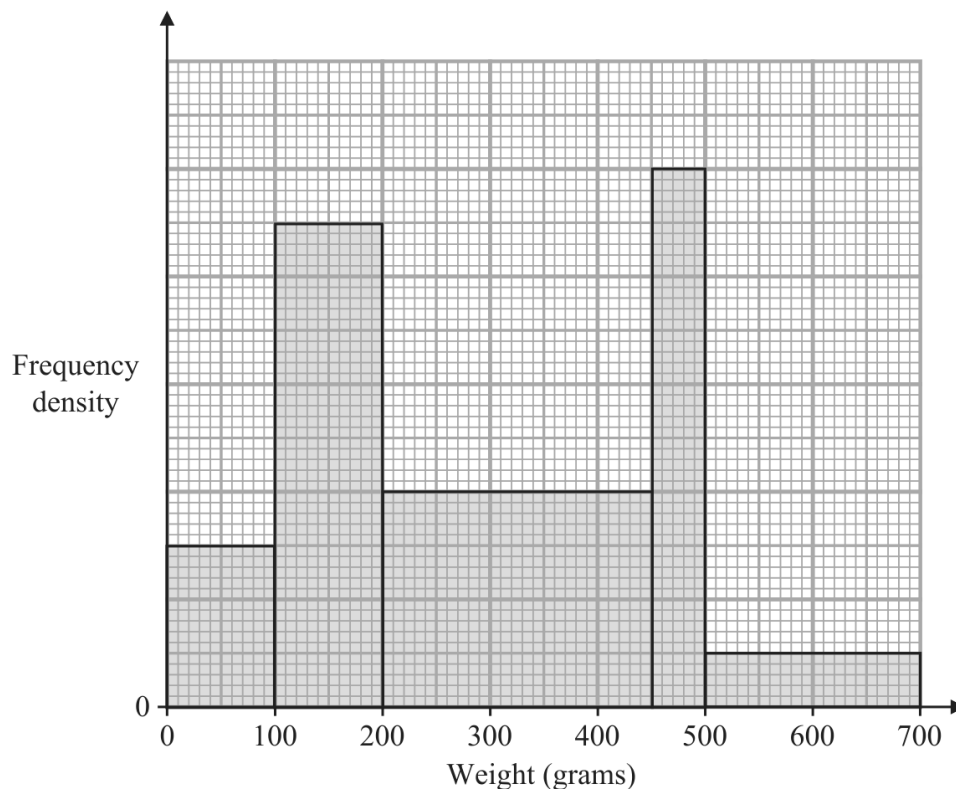
(Total for Question 17 is 3 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q20

CH 6 • TOPIC 2 4 marks

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

(Total for Question 20 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|--|
| 01 | Drawing | Width column; $fd = f \div w$; bars touch. |
| 02 | Reading | Area = frequency; known bar gives the scale. |
| 03 | Part of a bar | $fd \times$ part width; divide by the right total. |

BOTTOM LINE Height = fd • area = frequency • no scale → count squares from a known bar.

Answers

Histograms • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • FREQUENCY DENSITY AND DRAWING HISTOGRAMS

- Q1** Dr Eslam • Frequency density 1.2, 4, 0.8
- Q2** Practice • Medium Frequency densities 3.5, 7, 4.5, 3, 2.5
- Q3** Practice • Medium Frequency densities 0.75, 3.2, 4.2, 1.6
- Q4** Practice • Medium Frequency densities 5.4, 5, 2.2, 0.5
- Q5** Practice • Very Hard (a) fd 7, 8.4, 4, 3.6, 1.4, 0.55 (b) top 25% = 35: $11 + 14 = 25$, need 10 more of the 18 in $15 < a \leq 20$: about £17 (£17.22)
- Q6** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q18 Draw bars with densities 3.5, 3, 5, 2, 0.8, and 98 plants
- Q7** Edexcel 4MA1 • November 2021 • Paper 2H • Q14 Histogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$
- Q8** Edexcel 4MA1 • May 2024 • Paper 1HR • Q16 Draw bars with heights 2.5, 4, 3.6, 1.4, 0.6

IDEA 02 • READING A HISTOGRAM: AREA IS FREQUENCY

- Q9** Dr Eslam • Area $30 \times 1.4 = 42$
- Q10** Dr Eslam • Scale 1 small square = $\frac{20}{5 \times 8} = 0.5$ fd; $15 \times 2 = 30$ people
- Q11** Practice • Hard (a) 9, 8, 10, 12 (b) $\frac{8}{43}$ (c) 22nd value: about £26 000
- Q12** Practice • Hard $10 + 30 + 20 = 60$, so 14 in 170–180: bar height 1.4
- Q13** Practice • Hard Area of 50–55 bar = 44 out of 300: $\frac{44}{300} \times 150 = 22$
- Q14** Practice • Very Hard 28, 74, 36, 15; 77th value: $160 + \frac{49}{7.4} \approx 166.6$ cm < 172: the rowing club
- Q15** Edexcel 4MA1 • January 2020 • Paper 1HR • Q19 370 students
- Q16** Edexcel 4MA1 • January 2021 • Paper 1HR • Q18 4.36 hours
- Q17** Edexcel 4MA1 • May 2021 • Paper 1H • Q18 Completed frequencies 35, 40, 60, missing histogram density 1, and probability $\frac{12}{41}$
- Q18** Edexcel 4MA1 • January 2022 • Paper 1HR • Q21 102 watermelons
- Q19** Edexcel 4MA1 • June 2022 • Paper 1HR • Q19 14 trees
- Q20** Edexcel 4MA1 • June 2022 • Paper 1H • Q21 87 children
- Q21** Edexcel 4MA1 • May 2023 • Paper 1H • Q18 Missing frequencies 25, 60, and missing densities 4.6, 1.8, 0.7
- Q22** Edexcel 4MA1 • November 2023 • Paper 1H • Q17 About 42 medium oranges
- Q23** Edexcel 4MA1 • May 2024 • Paper 1H • Q22 Draw the missing bar from 5 to 15 with frequency density 2.7
- Q24** Edexcel 4MA1 • November 2025 • Paper 2H • Q21 39 trains

IDEA 03 • PART OF A BAR: PROPORTIONS AND PROBABILITY

- Q25** Dr Eslam • Part of a bar $\frac{5}{20} \times 30 = 7.5 \approx 8$
- Q26** Practice • Hard Total 600, more than an hour $54 + 18 = 72$:
12%
- Q27** Practice • Hard (i) $\frac{4+10+3}{30} = \frac{17}{30}$ (ii) scores spread evenly within
40–50
- Q28** Practice • Very Hard $\frac{76}{81} \approx 93.8\%$
- Q29** Practice • Very Hard $\frac{110}{620} = \frac{11}{62} \approx 0.177$
- Q30** Edexcel 4MA1 • January 2020 • Paper 2H • Q18 $\frac{23}{60}$, then $\frac{3}{5}$
- Q31** Edexcel 4MA1 • January 2021 • Paper 2H • Q15 142, then 0.201 approximately
- Q32** Edexcel 4MA1 • January 2022 • Paper 2H • Q20 $\frac{1}{6}$
- Q33** Edexcel 4MA1 • May 2023 • Paper 1HR • Q22 16.7%
- Q34** Edexcel 4MA1 • November 2024 • Paper 1H • Q18 $\frac{99}{145}$
- Q35** Edexcel 4MA1 • June 2025 • Paper 1H • Q17 $\frac{11}{63}$
- Q36** Edexcel 4MA1 • June 2025 • Paper 2HR • Q20 $\frac{8}{29}$ or 0.276 to 3 significant figures